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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Session	**2025-26 (Even)**	**Course Name**	**PE-III: Lab -Reinforcement Learning**
Semester	**6th**	**Course Code**	**23ADS1644**
Roll No	**42**	**Name of Student**	**Krishna Mishra**

Practical Number	**01**
Course Outcome	CO1: Apply fundamental reinforcement learning concepts to model and solve decision-making problems. CO2: Implement basic reinforcement learning algorithms to train agents in a given environment. CO3: Analyze agent behavior
Aim	To understand the Google Colab environment and various Python libraries used to import global dataset and analyze basic dataset characteristics.
Problem Definition	The objective of this experiment is to explore and understand the Google Colab environment and utilize Python libraries like Pandas and NumPy to import a dataset and perform basic data analysis. The task involves examining dataset structure, identifying missing values, and generating statistical summaries to gain insights into the dataset.
Theory (100 words)	Data analysis is a crucial step in data science that involves inspecting, cleaning, and transforming data to extract useful information. Python provides powerful libraries such as Pandas and NumPy for handling and analyzing datasets efficiently. Pandas offers data structures like DataFrames, which allow easy manipulation of structured data. Functions like head(), info(), and describe() help in understanding dataset characteristics, including data types, missing values, and statistical measures. Google Colab is a cloud-based platform that enables users to run Python code without local setup, making it ideal for data analysis tasks. By analyzing datasets such as the Iris dataset, users can learn how to explore data, identify patterns, and prepare it for further machine learning tasks.

Procedure and Execution (100 Words)	Algorithm: 1. Import required libraries: Pandas and NumPy. 2. Load the dataset (Iris.csv) using pd.read_csv(). 3. Display the first 5 rows using head() to preview data. 4. Use info() to get dataset structure and data types. 5. Check dataset dimensions using shape. 6. Display column names using columns. 7. Identify missing values using isnull().sum(). 8. Generate statistical summary using describe().

Code:

```
import pandas as pd
import numpy as np

url = "Iris.csv"
df = pd.read_csv(url)

print("First 5 rows of the dataset:")
display(df.head())

print("\nDataset Information:")
df.info()

print("\nDataset Shape (rows, columns):")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nMissing Values in Each Column:")
print(df.isnull().sum())

print("\nStatistical Summary:")
display(df.describe())
```

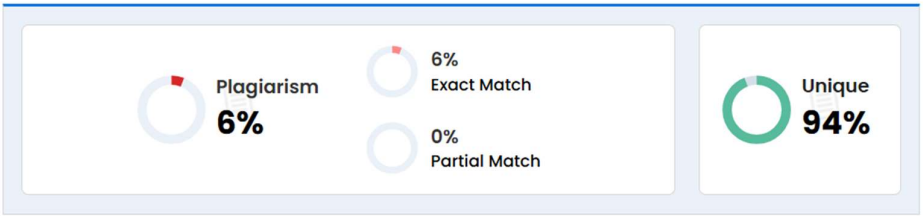
Output:

```
First 5 rows of the dataset:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB

Dataset Shape (rows, columns):
(150, 6)

Column Names:
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
       'Species'],
      dtype='object')

Missing Values in Each Column:
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64

Statistical Summary:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000

Output Analysis	• The dataset preview (head()) shows the first five records, helping understand the structure. • info() reveals data types (numerical or categorical) and confirms whether there are null values. • The shape indicates total rows and columns in the dataset. • Column names help identify features like sepal length, petal width, etc. • Missing value check ensures data quality; ideally, the Iris dataset has no missing values. • The statistical summary (describe()) provides mean, standard deviation, minimum, and maximum values, helping understand data distribution.
Link of student Github profile where lab assignment has been uploaded	https://github.com/krish-na45/RL_Practicals
Conclusion	This experiment successfully demonstrates how to use Google Colab and Python libraries for basic dataset exploration. It helps in understanding dataset structure, detecting missing values, and summarizing data statistically. Such analysis is essential as a first step before applying machine learning techniques, ensuring the data is clean and well-understood.
Plag Report (Similarity index < 12%)	
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